

Graph-theoretic approach to synchronizing stochastic coupled systems with time-varying delays on networks via periodically intermittent control



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ABSTRACT

This article concerns the inner exponential synchronization (IES) problem of a class of stochastic coupled systems with time-varying delays on networks (SCSTDNs). Periodically intermittent control is imposed on SCSTDNs to realize IES. By combining graph theory with Lyapunov method, a Lyapunov function is constructed and some sufficient conditions guaranteeing IES of SCSTDNs under periodically intermittent control are derived. Additionally, the analysis of IES for a stochastic coupled oscillators model with time-varying delays is performed to show the applicability of the analytical results. Finally, an example with numerical simulation is presented to illustrate the validity of our theoretical results.

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1. introduction

Over the past decades, coupled system on networks (CSNs), as a special kind of complex networks, have been studied extensively because they have many important applications in various areas such as moving image processing, optimization, speed detection of moving subjects, secure communication and so on, see [1–4] and references therein. Especially, the synchronization phenomena in CSNs have secured growing attention among the research community since the fact that synchronization can well explain many natural phenomena and also has lots of potential applications in chemical reactions, biological systems, information science and so on [5–7]. As a result, various kinds of synchronization problem of CSNs have been studied and numerous results has been proposed [8,9]. For instance, in [9], tang et al. discussed pin cluster synchronization on complex dynamical networks in Lur'e Forms. It should be pointed out that synchronization of all the nodes within a network occupies a pivotal and vital position in concrete application. Therefore, rapidly growing research interests have focused on the synchronization problem of CSNs, and a wide variety of synchronization criteria have been presented [10–12].

In general, control techniques have great influence on the realization of network synchronization. Hitherto, various control schemes have been used to study the synchronization of CSNs, such as state feedback control [13], impulsive control [14], pinning control [15] and intermittent control [16]. From the viewpoint of engineering applications, the control cost of continuous feedback control is expensive. Compared with continuous feedback control of synchronization, impulsive control and intermittent control are discontinuous control methods. The main idea of impulsive control is to change the states of continuous dynamic systems at certain time moments, while the intermittent control is to control systems via discontinuous

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control inputs at a time duration which is usually periodic. Hence, they can dramatically reduce the control cost and the amount of the transmitted information. Additionally, since the states of controlled systems are changed directly, impulsive control is an effective approach when the states are observable. But it seems to be invalid when the states of controlled systems are unobservable [16]. What's more, in comparison with impulsive control, periodically intermittent control is easier to be implemented in practice due to it has a nonzero control time. Indeed, intermittent control arises naturally in a wide variety of applications, such as ecosystem management, windshield wipers intermittent control, and control of hyper uricacidemia in the treatment of gout. Owing to those merits, periodically intermittent control have been applied successfully to study the synchronization problem of various kinds of coupled networks [17–19]. For instance, in [17], the synchronization problem for a class of complex delayed dynamical networks was investigated by pinning periodically intermittent control. In [19], some exponential synchronization criteria for coupled memristor-based chaotic neural networks with time-varying delays were derived via periodically intermittent control.

In real applications, CSNs are inevitably subject to various types of environmental noises which exist in external environment. Hence, one should analyze a more generalized model by considering the stochastic phenomena [20–22]. In [21], Yan et al. successfully studied the quantitative exponential stability and stabilisation problems of discrete-time Markov jump systems with multiplicative noises. On the other hand, due to the finite speed of information transmission and traffic congestions among the nodes, the synchronization problem of CSNs with coupling delays has widely been studied in the literature. For example, the synchronization problem of hybrid coupled complex dynamical networks with mixed time delays was discussed in [23]. Hence, the synchronization problem of various types of networks with environment noise and time delays has attracted much attention from researchers in different fields [24–26]. In [25], the problem of stochastic synchronization analysis was investigated for a new array of coupled discrete-time stochastic complex networks with randomly occurred nonlinearities and time delays. Synchronization and state estimation problems were considered for an array of coupled discrete time-varying stochastic complex networks over a finite horizon in [26]. Therefore, to make model more general, it is necessary to take stochastic disturbance and time delays into account.

It is a meaningful work to study the synchronization of CSNs, but it is also formidable. On the one hand, some factors dramatically rise the complexity of the systems. It is known that time delays and environmental noises are often sources of chaos of CSNs. Moreover, synchronization of CSNs is also affected by the complex coupling structure. On the other hand, the main method for dealing to the synchronization problem of stochastic coupled systems with time-varying delays on networks (SCSTDNs) is essentially based on the search of common Lyapunov functions or variations of the same framework. However, it is very difficult to construct a proper Lyapunov function for SCSTDNs directly, this is also a disadvantage of the Lyapunov approach. Therefore, the synchronization analysis for SCSTDNs is quite involved and complex. It is inspiring that Michael Y. Li and his co-authors [1] developed a systematic approach that took advantage of graph theory to construct global Lyapunov functions when investigating stability of large-scale coupled ordinary differential systems. The graph-theoretical method can avoid the difficulty in directly constructing global Lyapunov functions for coupled systems. Lots of literature which indicate the graph-theoretical approach is applicable in kinds of coupled networks have been published and the graph-theoretical approach has been regarded as an important method for dynamical analysis of CSNs [27–31]. In [27], stability of a class of stochastic coupled systems with Lévy noise on networks was studied by using this graph-theoretical approach. Synchronization of stochastic coupled systems with feedback control based on discrete-time state observations was investigated via the graph-theoretical method in [28]. Although there have been many results on the problem of dynamical analysis for CSNs, there are nearly no results on the inner exponential synchronization (IES) issue of SCSTDNs under periodically intermittent control by utilizing the graph-theoretical technique.

Motivated by the aforementioned discussion, the aim of this paper is to study the IES problem of SCSTDNs under periodically intermittent control. The contributions are as follows. Firstly, independent time-varying delays influence the state of different vertices without the disturbance of white noise and influence the intension of white noise, respectively, are all taken into account. Secondly, by combing graph theory with Lyapunov method, a Lyapunov-type theorem is obtained to ensure IES of SCSTDNs under periodically intermittent control. After that, a coefficient-type theorem in terms of the coefficients of the formulated systems is derived on the basis of the Lyapunov-type theorem. Thirdly, we apply our theoretic results to stochastic coupled oscillators with time-varying delays to show the applicability of our main results. Finally, a numerical simulation is given to illustrate the validity of the proposed criteria.

The reminder of this paper are as follows. In Section 2, some preliminaries and a mathematical model of SCSTDNs are presented. Some sufficient conditions to ensure IES of SCSTDNs under periodically intermittent control are established in Section 3. In Section 4, we apply our theoretic results to stochastic coupled oscillators with time-varying delays to show the practicality of the theoretic results. A numerical example is provided in Section 5. Conclusion is finally drawn in Section 6.

2. Preliminaries and model formulation

In this section, we first introduce some basic notations and a vital lemma in graph theory. Then, we give the model formulation and finally present two important definitions.

2.1. Preliminaries

Throughout this paper, we let $\mathbb{R}^m$ stand for m -dimensional Euclidean space and if $x \in \mathbb{R}^n$, then $|x|$ is Euclidean norm. The notations $\mathbb{L} = \{1, 2, \dots, n\}$, $\mathbb{Z}^+ = \{1, 2, \dots\}$ and $\mathbb{R}^+ = [0, +\infty)$ are used. Let superscript “T” denote the transpose of a vector or matrix. Write $C([-\tau, 0]; \mathbb{R}^n)$ for the family of all continuous functions from $[-\tau, 0]$ to $\mathbb{R}^n$ with norm $\|x\| = \sup_{s \in [-\tau, 0]} |x(s)|$. Denote $C^{2,1}(\mathbb{R}^m \times \mathbb{R}^+; \mathbb{R}^+)$ as the family of all nonnegative functions $V(x, t)$ on $\mathbb{R}^m \times \mathbb{R}^+$ which are continuously twice differentiable in x and once in t . Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space with a filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions (i.e., it is right continuous and $\mathcal{F}_0$ contains all $\mathbb{P}$ -null sets) and $L^2_{\mathcal{F}_0}([-\tau, 0], \mathbb{R}^{(n-1)m})$ stand for the family of $\mathcal{F}_0$ -measurable $C([-\tau, 0], \mathbb{R}^{(n-1)m})$ -value random variable x such that $\mathbb{E}\|x\|^2 < \infty$. Let $W(t)$ stand for the white noise defined on the probability space. The mathematical expectation with respect to the given probability measure $\mathbb{P}$ is denoted by $\mathbb{E}\{\cdot\}$. Given a weighted digraph $\mathcal{G}$ with n vertices, the weight matrix is denoted by $A = (a_{ij})_{n \times n}$ whose entry a_{ij} equals the weighted of arc (j, i) . And P is the Laplacian matrix of $(\mathcal{G}, A)$, in which P is defined as $P = (p_{kl})_{n \times n}$, where $p_{kl} = -a_{kl}$ for $k \neq l$ and $p_{kl} = \sum_{j \neq k} a_{kj}$ for $k = l$.

Now we give an important lemma in graph theory. Other basic concepts of graph theory involved in this paper can referred to in [1] and [35].

Lemma 1. (Li [1]) Assume $n \geq 2$. Let $c^{(i)}$ denote the cofactor of the i th diagonal element of P which stands for the Laplacian matrix of $(\mathcal{G}, A)$. Then the following identity holds:

$$\sum_{i,j=1}^n c^{(i)} a_{ij} F_{ij}(x_i, x_j) = \sum_{Q \in \mathcal{Q}} W(Q) \sum_{(s,r) \in E(C_Q)} F_{rs}(x_r, x_s),$$

where $F_{ij}(x_i, x_j) : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ is an arbitrary function for any $i, j \in \mathbb{L}$, $\mathcal{Q}$ denotes the set of all spanning unicyclic graphs of digraph $(\mathcal{G}, A)$, $W(Q)$ is the weight of Q and C_Q denotes the directed cycle of Q . Particularly if $(\mathcal{G}, A)$ is strongly connected, $c^{(i)} > 0$ for any $i \in \mathbb{L}$.

2.2. Model formulation

Based on a digraph $\mathcal{G}$ with n ($n \geq 2$) vertices, we consider the following stochastic coupled systems with time-varying delays:

$$\begin{aligned} dx_i(t) = & \left(f(x_i(t), t) + \sum_{k=1}^n a_{ik} H_{ik}(x_i(t), x_k(t)) + \sum_{k=1}^n b_{ik} M_{ik}(x_i(t - \tau_1(t)), x_k(t - \tau_1(t))) + I(t) \right) dt \\ & + \left(g(x_i(t), t) + \sum_{k=1}^n c_{ik} N_{ik}(x_i(t - \tau_2(t)), x_k(t - \tau_2(t))) \right) dW(t), \quad t \geq 0, \quad i \in \mathbb{L}, \end{aligned} \quad (1)$$

where $x_i(t) \in \mathbb{R}^m$ is the state vector of the i th vertex system; $f: \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}^m$ is a continuous vector-valued function; $g: \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}^m$ stands for the perturbation intense of white noise; M_{ik} and $H_{ik} : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}^m$, respectively, denote the influence of the state of the k th vertex on the i th vertex with or without delay; $N_{ik} : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ represents the influence of white noise of the k th vertex on the i th vertex with time delay; the strength of coupling without time delay is represented by a_{ik} ; b_{ik} and c_{ik} represent the strengths of delayed coupling. Here, $a_{ik} \geq 0$, $b_{ik} \geq 0$ and $c_{ik} \geq 0$, specially, $a_{ik} = 0$, $b_{ik} = 0$, $c_{ik} = 0$ means that there exists no arc from the k th vertex to the i th vertex; the coupling delay functions $\tau_1(t)$ and $\tau_2(t)$ satisfy $0 \leq \tau_1(t) \leq \tau_1$ and $0 \leq \tau_2(t) \leq \tau_2$, where τ_1 and τ_2 are known constants; and $I(t)$ is the external input.

In order to synchronize system (1), for any fixed $i \in \mathbb{L}$, periodically intermittent control is added to the j th ($j \in \mathbb{L}$, $j \neq i$) vertex system, then the j th vertex system with periodically intermittent control can be described by

$$\begin{aligned} dx_j(t) = & \left(f(x_j(t), t) + \sum_{k=1}^n a_{jk} H_{jk}(x_j(t), x_k(t)) + \sum_{k=1}^n b_{jk} M_{jk}(x_j(t - \tau_1(t)), x_k(t - \tau_1(t))) + I(t) + u_{ij}(t) \right) dt \\ & + \left(g(x_j(t), t) + \sum_{k=1}^n c_{jk} N_{jk}(x_j(t - \tau_2(t)), x_k(t - \tau_2(t))) \right) dW(t), \quad t \geq 0, \end{aligned} \quad (2)$$

where $u_{ij}(t)$ is periodically intermittent control defined by

$$u_{ij}(t) = \begin{cases} -k_{ij}(x_j(t) - x_i(t)), & lT \leq t < (l + \theta)T, \\ 0, & (l + \theta)T \leq t < (l + 1)T, \end{cases} \quad (3)$$

in which $l \in \mathbb{Z}^+$, $k_{ij} > 0$ denotes the constant control strength, T stands for the control period and $0 < \theta < 1$ is the rate of control duration.

Furthermore, let $x_{ij}(t) = x_j(t) - x_i(t)$ be the synchronization error of system (1), the dynamical system of inner synchronization error with control (3) is written as

$$\begin{aligned} dx_{ij}(t) = & \left(f(x_j(t), t) - f(x_i(t), t) + \sum_{k=1}^n a_{jk} H_{jk}(x_j(t), x_k(t)) - \sum_{k=1}^n a_{ik} H_{ik}(x_i(t), x_k(t)) \right. \\ & + \sum_{k=1}^n b_{jk} M_{jk}(x_j(t - \tau_1(t)), x_k(t - \tau_1(t))) - \sum_{k=1}^n b_{ik} M_{ik}(x_i(t - \tau_1(t)), x_k(t - \tau_1(t))) + u_{ij}(t) \Big) dt \\ & + \left(g(x_j(t), t) - g(x_i(t), t) + \sum_{k=1}^n c_{jk} N_{jk}(x_j(t - \tau_2(t)), x_k(t - \tau_2(t))) \right. \\ & \left. - \sum_{k=1}^n c_{ik} N_{ik}(x_i(t - \tau_2(t)), x_k(t - \tau_2(t))) \right) dW(t), \quad t \geq 0, \quad i, j \in \mathbb{L}, \quad j \neq i. \end{aligned} \quad (4)$$

Assume that the coefficients in system (4) satisfy the usually local Lipschitz condition and linear growth condition. Hence, for any given initial value $\psi \in L^2_{\mathcal{F}_0}([-\tau, 0], \mathbb{R}^{(n-1)m})$, where $\tau = \max\{\tau_1, \tau_2\}$, the existence of a unique solution $x(t) = (x_{i1}^T(t), x_{i2}^T(t), \dots, x_{i(n-1)}^T(t), x_{in}^T(t), \dots, x_{in}^T(t))^T$ to system (4) can be guaranteed by [36]. In addition, the vector functions in system (4) satisfy $f(0, t) = 0$, $H_{ik}(0, 0) = 0$, $H_{jk}(0, 0) = 0$, $M_{ik}(0, 0) = 0$, $M_{jk}(0, 0) = 0$, $g(0, t) = 0$, $N_{ik}(0, 0) = 0$ and $N_{jk}(0, 0) = 0$, which means system (4) under periodically intermittent control (3) has a trivial solution $x(t) = 0$.

Remark 1. Numerous researchers have worked on dynamical analysis of complex networks with time delays or stochastic disturbances in the past decades. In [28], synchronization of stochastic coupled systems via feedback control based on discrete-time state observations was investigated. Therein, the stochastic coupled systems did not consider time delays. The environment noise was not been taken into account in [31]. In this paper, we take both environment noise and time delays into consideration to further study the synchronization issue of system (1). Additionally, there are few results of the situation in which the stochastic disturbance is affected by the time delays. The dynamical analysis for stochastic networks of coupled oscillators with variable delay was investigated in [34]. Unfortunately, the time delays in the coupling state without the disturbance of white noises and in the coupling state with the disturbance of white noises are assumed to be the same. Clearly, in a real-word situation, they should be different from each other. In this paper, a more general case where the coupling state and the white noise are affected by independent time-varying delays.

Throughout the paper, we make the following hypotheses:

(H1.) For any $i, k \in \mathbb{L}$, there are positive numbers P_{ik} , Q_{ik} and V_{ik} such that

$$|H_{ik}(x_i, x_k)| \leq P_{ik} |x_i - x_k|, \quad (5)$$

$$|M_{ik}(x_i, x_k)| \leq Q_{ik} |x_i - x_k|, \quad (6)$$

$$|N_{ik}(x_i, x_k)| \leq V_{ik} |x_i - x_k|. \quad (7)$$

(H2.) For any $i, j \in \mathbb{L}$, there exists a positive scaler δ such that

$$|g(x_i, t) - g(x_j, t)| \leq \delta |x_i - x_j|. \quad (8)$$

For the sake of simplicity, for any $i, k \in \mathbb{L}$ we denote

$$H_{ik} = H_{ik}(x_i(t), x_k(t)), \quad M_{ik} = M_{ik}(x_i(t - \tau_1(t)), x_k(t - \tau_1(t))), \quad N_{ik} = N_{ik}(x_i(t - \tau_2(t)), x_k(t - \tau_2(t))).$$

In what follows, we introduce the definitions of a differential operator and IES for system (1), which will be utilized to obtain our main results.

Definition 1 [35]. For $V^{ij}(x_{ij}(t), t) \in C^{2,1}(\mathbb{R}^m \times \mathbb{R}^+; \mathbb{R}^+)$, along with the solution of system (1), the differential operator $\mathcal{L}V^{ij}(x_{ij}(t), t)$ is defined by

$$\begin{aligned} \mathcal{L}V^{ij}(x_{ij}(t), t) &= \frac{\partial V^{ij}(x_{ij}(t), t)}{\partial t} + \frac{\partial V^{ij}(x_{ij}(t), t)}{\partial x_{ij}(t)} \left(f(x_j(t), t) - f(x_i(t), t) + \sum_{k=1}^n a_{jk} H_{jk} - \sum_{k=1}^n a_{ik} H_{ik} + \sum_{k=1}^n b_{jk} M_{jk} - \sum_{k=1}^n b_{ik} M_{ik} + u_{ij}(t) \right) \\ &+ \frac{1}{2} \left(g(x_j(t), t) - g(x_i(t), t) + \sum_{k=1}^n c_{jk} N_{jk} - \sum_{k=1}^n c_{ik} N_{ik} \right)^T \frac{\partial^2 V^{ij}(x_{ij}(t), t)}{\partial x_{ij}^2(t)} \\ &\times \left(g(x_j(t), t) - g(x_i(t), t) + \sum_{k=1}^n c_{jk} N_{jk} - \sum_{k=1}^n c_{ik} N_{ik} \right), \end{aligned} \quad (9)$$

in which

$$\frac{\partial V^{ij}(x_{ij}(t), t)}{\partial x_{ij}(t)} = \left(\frac{\partial V^{ij}(x_{ij}(t), t)}{\partial x_{ij}^{(1)}(t)}, \dots, \frac{\partial V^{ij}(x_{ij}(t), t)}{\partial x_{ij}^{(m)}(t)} \right), \quad \frac{\partial^2 V^{ij}(x_{ij}(t), t)}{\partial x_{ij}^2(t)} = \left(\frac{\partial^2 V^{ij}(x_{ij}(t), t)}{\partial x_{ij}^{(k)}(t) \partial x_{ij}^{(h)}(t)} \right)_{m \times m}.$$

Definition 2 [19]. System (1) is said to be inner exponentially synchronized, if for any initial condition ψ , there exist $M > 0$ and $\lambda > 0$, such that

$$\mathbb{E}|x_i(t) - x_j(t)|^2 \leq M e^{-\lambda t} \mathbb{E}\|\psi\|^2$$

for any $t \geq 0$ and $i, j \in \mathbb{L}$. Here λ is called the degree of exponential synchronization.

3. Main results

In this section, we consider the IES problem of system (1) under periodically intermittent control (3). Some IES criteria for system (1) will be derived by combining Lyapunov method with graph theory.

3.1. Lyapunov-type theorem

In order to construct a Lyapunov function for system (4), a theoretical framework will be proposed and a Lyapunov-type theorem will be provided in this section. For a clear presentation, throughout the article we denote

$$x_{ij}(\cdot) = x_j(\cdot) - x_i(\cdot), \quad x_{ik}(\cdot) = x_i(\cdot) - x_k(\cdot), \quad x_{jk}(\cdot) = x_j(\cdot) - x_k(\cdot).$$

Definition 3. The set $\{V^{ij}(x_{ij}, t) \mid V^{ij}(x_{ij}, t) \in C^{2,1}(\mathbb{R}^m \times \mathbb{R}^+; \mathbb{R}^+)\}$ is called a vertex-Lyapunov functions set for system (4) if the following conditions hold for any fixed i ($i \in \mathbb{L}$) and any j ($j \in \mathbb{L}, j \neq i$):

A1. There are positive constants α_{ij} and β_{ij} , such that

$$\alpha_{ij}|x_{ij}|^2 \leq V^{ij}(x_{ij}, t) \leq \beta_{ij}|x_{ij}|^2.$$

A2. There exist positive constants $\sigma_{ij}^{(2)}$, μ_{ij} , ρ_{ij} , function $G_{ik}(X_{ik}(t)) \in C(\mathbb{R}^{2m} \times \mathbb{R}^{2m}; \mathbb{R})$ and a matrix $A = (h_{ijk})_{(n-1) \times (n-1)}$ satisfying

$$\mathcal{L}V^{ij}(x_{ij}(t), t) \leq \begin{cases} -\sigma_{ij}^{(1)}|x_{ij}(t)|^2 + \mu_{ij}|x_{ij}(t - \tau_1(t))|^2 + \rho_{ij}|x_{ij}(t - \tau_2(t))|^2 & lT \leq t < (l + \theta)T, \\ \quad + \sum_{k=1, k \neq i}^n h_{ijk} \left(G_{ik}(X_{ik}(t)) - G_{ij}(X_{ij}(t)) \right), \\ -\sigma_{ij}^{(2)}|x_{ij}(t)|^2 + \mu_{ij}|x_{ij}(t - \tau_1(t))|^2 + \rho_{ij}|x_{ij}(t - \tau_2(t))|^2 & (l + \theta)T \leq t < (l + 1)T, \\ \quad + \sum_{k=1, k \neq i}^n h_{ijk} \left(G_{ik}(X_{ik}(t)) - G_{ij}(X_{ij}(t)) \right), \end{cases}$$

in which $\sigma_{ij}^{(1)} = \sigma_{ij}^{(2)} + 2k_{ij}$ and $X_{ik}(t) = |x_{ik}(t)|^2 + |x_{ik}(t - \tau_1(t))|^2 + 2|x_{ik}(t - \tau_2(t))|^2$.

Theorem 1. Suppose that system (4) admits a vertex-Lyapunov functions set $\{V^{ij}(x_{ij}, t)\}$ and digraph (G, A) is strongly connected. If $\theta > 1 - \xi/\gamma$, where $\xi > 0$ is a constant and satisfies

$$-\xi + \min_{i, j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(1)}}{\beta_{ij}} \right\} - \max_{i, j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} e^{\xi\tau} - \max_{i, j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\} e^{\xi\tau} \geq 0, \quad (10)$$

where $\gamma = \max_{i, j \in \mathbb{L}} \{2k_{ij}/\beta_{ij}\}$. Then system (1) can realize IES with $\lambda = \xi - \gamma(1 - \theta)$ under periodically intermittent control (3).

Proof. Define

$$V(x, t) = \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} V^{ij}(x_{ij}, t),$$

where c_{ij} is the cofactor of the j th diagonal element of the Laplacian matrix of (G, A) for any fixed i ($i \in \mathbb{L}$). By Lemma 1, we have $c_{ij} > 0$ for any $j \in \mathbb{L}$.

Denote $\iota = \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \beta_{ij}$ and $M_0 = \iota \sup_{s \in [-\tau, 0]} |\psi(s)|^2$, when $t \in [-\tau, 0]$ from (3), we have

$$\mathbb{E}V(x, t) \leq \mathbb{E} \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \beta_{ij} |x_{ij}|^2 \leq \left(\sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \beta_{ij} \right) \mathbb{E}\|\psi\|^2 \leq \mathbb{E}M_0. \quad (11)$$

In order to convenient the reader, the proof is divided into the following two parts:

Part 1. Prove $\mathbb{E}P(t) = e^{\xi t} \mathbb{E}V(x(t), t) - h \mathbb{E}M_0 < 0$, for all $t \in [-\tau, \theta T]$, where $h > 1$ is a scalar.

Firstly, from (11), it is obvious that

$$\mathbb{E}P(t) < 0, \quad t \in [-\tau, 0]. \quad (12)$$

Next, we prove that

$$\mathbb{E}P(t) < 0, \quad t \in [0, \theta T]. \quad (13)$$

Otherwise, by (12) and the continuity of $V(x, t)$ as $t \in [-\tau, +\infty)$, there exists a $t_0 \in [0, \theta T]$, such that

$$\mathbb{E}P(t_0) = 0, \quad D^+\mathbb{E}P(t_0) \geq 0, \quad (14)$$

and

$$\mathbb{E}P(t) < 0, \quad \text{for } t \in [0, t_0]. \quad (15)$$

If necessary, one can introduce the stopping times. Here, by making use of condition A2 and Lemma 1, we can see that

$$\begin{aligned} D^+\mathbb{E}P(t_0) &= \xi e^{\xi t_0} \mathbb{E}V(x(t_0), t_0) + e^{\xi t_0} D^+\mathbb{E}V(x(t_0), t_0) \\ &\leq e^{\xi t_0} \mathbb{E} \left\{ \xi V(x(t_0), t_0) + \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \left(-\sigma_{ij}^{(1)} |x_{ij}(t_0)|^2 + \mu_{ij} |x_{ij}(t_0 - \tau_1(t_0))|^2 + \rho_{ij} |x_{ij}(t_0 - \tau_2(t_0))|^2 \right) \right. \\ &\quad \left. + \sum_{i=1}^n \sum_{j=1, j \neq i}^n \sum_{k=1, k \neq i}^n c_{ij} h_{ijk} \left(G_{ik}(X_{ik}(t_0)) - G_{ij}(X_{ij}(t_0)) \right) \right\} \\ &\leq e^{\xi t_0} \mathbb{E} \left\{ \xi V(x(t_0), t_0) - \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \frac{\sigma_{ij}^{(1)}}{\beta_{ij}} V^{ij}(x_{ij}(t_0), t_0) + \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \frac{\mu_{ij}}{\alpha_{ij}} V^{ij}(x_{ij}(t_0 - \tau_1(t_0)), t_0 - \tau_1(t_0)) \right. \\ &\quad \left. + \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \frac{\rho_{ij}}{\alpha_{ij}} V^{ij}(x_{ij}(t_0 - \tau_2(t_0)), t_0 - \tau_2(t_0)) + \sum_{i=1}^n \sum_{Q \in \mathcal{Q}} W(Q) \sum_{(s,r) \in E(C_Q)} \left(G_{ir}(X_{ir}(t_0)) - G_{is}(X_{is}(t_0)) \right) \right\}. \end{aligned} \quad (16)$$

Along each directed cycle $\mathcal{C}$, there is

$$\sum_{(s,r) \in E(C_Q)} \left(G_{ir}(X_{ir}(t_0)) - G_{is}(X_{is}(t_0)) \right) = 0. \quad (17)$$

Inserting (14), (15) and (17) into (16), we get

$$\begin{aligned} D^+\mathbb{E}P(t_0) &\leq \mathbb{E} \left\{ \left(\xi - \min_{i,j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(1)}}{\beta_{ij}} \right\} \right) e^{\xi t_0} V(x(t_0), t_0) + \max_{i,j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} e^{\xi t_0} V(x(t_0 - \tau_1(t_0)), t_0 - \tau_1(t_0)) \right. \\ &\quad \left. + \max_{i,j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\} e^{\xi t_0} V(x(t_0 - \tau_2(t_0)), t_0 - \tau_2(t_0)) \right\} \\ &< \left(\xi - \min_{i,j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(1)}}{\beta_{ij}} \right\} \right) hM_0 + \max_{i,j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} hM_0 + \max_{i,j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} h\mathbb{E}M_0 \\ &= - \left(-\xi + \min_{i,j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(1)}}{\beta_{ij}} \right\} - \max_{i,j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} - \max_{i,j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} \right) h\mathbb{E}M_0. \end{aligned}$$

From (10), one has $D^+\mathbb{E}P(t_0) < 0$. This contradicts $D^+\mathbb{E}P(t_0) \geq 0$, which implies that (13) holds.

Part 2. Prove $\mathbb{E}Q(t) = e^{\xi t} \mathbb{E}V(x(t), t) - h\mathbb{E}M_0 e^{\gamma(t-\theta T)} < 0$, for all $t \in [\theta T, T]$.

Suppose that the inequality $\mathbb{E}Q(t) < 0$ is false for all $t \in [\theta T, T]$, that is to say, there exists a point $t_1 \in [\theta T, T]$ such that

$$\mathbb{E}Q(t_1) = 0, \quad D^+\mathbb{E}Q(t_1) \geq 0 \quad (18)$$

and

$$\mathbb{E}Q(t) < 0, \quad \text{for } t \in [\theta T, t_1]. \quad (19)$$

If $\theta T \leq t_1 - \tau_1(t_1) \leq t_1$, it is easy to see that

$$e^{\xi t_1} \mathbb{E}V(x(t_1 - \tau_1(t_1)), t_1 - \tau_1(t_1)) < e^{\xi \tau} h\mathbb{E}M_0 e^{\gamma(t_1 - \theta T)}.$$

From Part 1, if $-\tau \leq t_1 - \tau_1(t_1) \leq \theta T$, we get

$$e^{\xi t_1} \mathbb{E}V(x(t_1 - \tau_1(t_1)), t_1 - \tau_1(t_1)) < e^{\xi \tau} h\mathbb{E}M_0 < e^{\xi \tau} h\mathbb{E}M_0 e^{\gamma(t_1 - \theta T)}.$$

Then, for $\tau_1(t_1) > 0$ the following inequality always holds:

$$e^{\xi t_1} \mathbb{E}V(x(t_1 - \tau_1(t_1)), t_1 - \tau_1(t_1)) < e^{\xi \tau} h \mathbb{E}M_0 e^{\gamma(t_1 - \theta T)}. \quad (20)$$

Similarly, for $\tau_2(t_1) > 0$ we can find that

$$e^{\xi t_1} \mathbb{E}V(x(t_1 - \tau_2(t_1)), t_1 - \tau_2(t_1)) < e^{\xi \tau} h \mathbb{E}M_0 e^{\gamma(t_1 - \theta T)}. \quad (21)$$

Using Lemma 1, condition A2, (10), (18)–(21), we have

$$\begin{aligned} D^+ \mathbb{E}Q(t_1) &= \xi e^{\xi t_1} \mathbb{E} \left\{ V(x(t_1), t_1) + e^{\xi t_1} D^+ V(x(t_1), t_1) - \gamma h M_0 e^{\gamma(t_1 - \theta T)} \right\} \\ &\leq e^{\xi t_1} \mathbb{E} \left\{ \xi V(x(t_1), t_1) - \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \sigma_{ij}^{(2)} |x_{ij}(t_1)|^2 + \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \mu_{ij} |x_{ij}(t_1 - \tau_1(t_1))|^2 \right. \\ &\quad \left. + \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} \rho_{ij} |x_{ij}(t_1 - \tau_2(t_1))|^2 - \gamma h M_0 e^{\gamma(t_1 - \theta T)} + e^{\xi t_1} \sum_{i=1}^n \sum_{j=1, j \neq i}^n \sum_{k=1, k \neq i}^n c_{ij} h_{ijk} (G_{ik}(X_{ik}(t_1)) - G_{ij}(X_{ij}(t_1))) \right\} \\ &\leq e^{\xi t_1} \mathbb{E} \left\{ \xi V(x(t_1), t_1) - \min_{i,j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(2)}}{\beta_{ij}} \right\} V(x(t_1), t_1) + \max_{i,j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} V(x(t_1 - \tau_1(t_1)), t_1 - \tau_1(t_1)) \right. \\ &\quad \left. + \max_{i,j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\} V(x(t_1 - \tau_2(t_1)), t_1 - \tau_2(t_1)) - \gamma h M_0 e^{\gamma(t_1 - \theta T)} \right\} \\ &< -\mathbb{E} \left\{ \left(-\xi + \min_{i,j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(2)}}{\beta_{ij}} \right\} - \max_{i,j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} - \max_{i,j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} \right) h M_0 e^{\gamma(t_1 - \theta T)} + \gamma h M_0 e^{\gamma(t_1 - \theta T)} \right\} \\ &\leq -\mathbb{E} \left\{ \left(-\xi + \min_{i,j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(1)}}{\beta_{ij}} \right\} - \max_{i,j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} - \max_{i,j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} \right) h M_0 e^{\gamma(t_1 - \theta T)} \right. \\ &\quad \left. - \left(\max_{i,j \in \mathbb{L}} \left\{ \frac{2k_{ij}}{\beta_{ij}} \right\} - \gamma \right) h M_0 e^{\gamma(t_1 - \theta T)} \right\} \\ &\leq 0, \end{aligned}$$

which contradicts $D^+ \mathbb{E}Q(t_1) \geq 0$. Therefore, $\mathbb{E}Q(t) < 0$, which indicates that

$$e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 e^{\gamma(t - \theta T)}.$$

As for $t \in [-\tau, \theta T]$, one has $e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 < h \mathbb{E}M_0 e^{\gamma(1 - \theta)T}$.

Thus, we can get

$$e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 e^{\gamma(1 - \theta)T}, \quad \text{for all } t \in [-\tau, T].$$

Similar to the proof of Part 1, for $T \leq t < (1 + \theta)T$, we have

$$e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 e^{\gamma(1 - \theta)T},$$

and for $(1 + \theta)T \leq t < 2T$

$$e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 e^{\gamma(1 - \theta)T} e^{\gamma[t - (1 + \theta)T]} = h \mathbb{E}M_0 e^{\gamma(t - 2\theta T)}.$$

By utilizing mathematical induction method, the following estimates are obtained for any integer l . For all $lT \leq t < (l + \theta)T$:

$$e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 e^{l\gamma(1 - \theta)T}, \quad (22)$$

and for $(l + \theta)T \leq t < (l + 1)T$,

$$e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 e^{\gamma(1 - \theta)T} e^{\gamma[t - (l + \theta)T]} = h \mathbb{E}M_0 e^{\gamma[t - (l + 1)\theta T]}. \quad (23)$$

Since for any $t \geq 0$, there exists a nonnegative integer l such that $lT \leq t < (l + \theta)T$, the following estimation for any t by (22) and (23) are proposed:

When $lT \leq t < (l + \theta)T$, and $l \leq t/T$, we can get

$$e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 e^{n\gamma(1 - \theta)T} \leq h \mathbb{E}M_0 e^{\gamma(1 - \theta)t}.$$

Meanwhile, when $(l + \theta)T \leq t < (l + 1)T$, and $l + 1 > t/T$, we can obtain

$$e^{\xi t} \mathbb{E}V(x(t), t) < h \mathbb{E}M_0 e^{\gamma(1 - \theta)t}.$$

Furthermore, let $h \rightarrow 1$, according to the definition of $V(x(t), t)$, it is readily seen that

$$\mathbb{E} \left(\sum_{i=1}^l \sum_{j=1, j \neq i}^l c_{ij} \alpha_{ij} |x_{ij}(t)|^2 \right) \leq \mathbb{E} V(x(t), t) \leq \mathbb{E} M_0 e^{-\xi t} e^{\gamma(1-\theta)t}.$$

Let $\lambda_0 = \min_{i,j \in \mathbb{L}} \{c_{ij} \alpha_{ij}\}$, we can derive

$$\mathbb{E} \left(\sum_{i=1}^l \sum_{j=1, j \neq i}^l |x_{ij}(t)|^2 \right) \leq \left(\frac{l}{\lambda_0} \right) e^{-[\xi - \gamma(1-\theta)]t} \mathbb{E} \|\psi\|^2 = M e^{-\lambda t} \mathbb{E} \|\psi\|^2, \quad t \geq 0,$$

where $\lambda = \xi - \gamma(1 - \theta) > 0$ and $M = l/\lambda_0 \geq 0$. Therefore, IES of system (1) can be realized via periodically intermittent control (3). This completes the proof. $\square$

Remark 2. Obviously, if

$$\min_{i,j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(1)}}{\beta_{ij}} \right\} > \max_{i,j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} + \max_{i,j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\},$$

there exist a scalar ξ such that

$$-\xi + \min_{i,j \in \mathbb{L}} \left\{ \frac{\sigma_{ij}^{(1)}}{\beta_{ij}} \right\} - \max_{i,j \in \mathbb{L}} \left\{ \frac{\mu_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} - \max_{i,j \in \mathbb{L}} \left\{ \frac{\rho_{ij}}{\alpha_{ij}} \right\} e^{\xi \tau} \geq 0.$$

It can be seen that the parameter ξ is relative to the parameter of $V^{ij}(x_{ij}(t), t)$ closely. What is more, after choosing the parameter ξ and control strength k_{ij} , the scope of control duration can be obtained by $\theta > 1 - \xi/\gamma$. In this way, we could design the time interval of the periodically intermittent control (3).

Remark 3. Noting that, the convergence speed of synchronization is an important feature of the performance of the controller. In this paper, we can accelerate the convergence speed of IES for system (1) by increasing θ , since the convergence speed of IES is $\lambda = \xi - \gamma(1 - \theta)$, which is closely related to the control strength k_{ij} and the parameter ξ . Hence, one can get desired convergence speed of IES for system (1) by choosing k_{ij} and ξ appropriately.

Remark 4. As is known to all, Lyapunov-Krasovskii functional method have been widely used to study the synchronization problem of delayed complex networks [32,33]. However, it is very difficult to obtain sufficient and necessary conditions by adopting Lyapunov-Krasovskii functional method to discuss the problem of synchronization for delayed complex networks. And furthermore it is also a formidable task to construct a proper Lyapunov functional for a given system. The feature of this paper is that SCSTDNs investigated in this article are described as a digraph $\mathcal{G}$, in which every subsystem is viewed as a vertex system, while the coupling relationship is shown by directed arcs with weight. A global Lyapunov function $V(x, t)$ of SCSTDNs is successfully constructed by taking advantage of graph theory, that is, $V(x, t) = \sum_{i=1}^n \sum_{j=1, j \neq i}^n c_{ij} V^{ij}(x_{ij}, t)$, which implies that the difficulty of constructing global Lyapunov functions directly is overcome with the help of some results in graph theory. Hence, synchronization for SCSTDNs will be studied much easier.

3.2. Coefficient-type theorem

In this subsection, a coefficient-type theorem will be provided to determine IES of system (1) on the basis of Theorem 1. The coefficient-type theorem is convenient to use in practice since it can be verified directly by the coefficients in system (1).

Theorem 2. Suppose that digraph $(\mathcal{G}, A)$ is strongly connected and H1, H2, Assumption 1 maintain. System (1) can achieve IES under periodically intermittent control (3) provided that the following conditions hold:

B1. There exists a positive constant v_{ij} , such that

$$x_{ij}^T F(x_{ij}, t) \leq -v_{ij} |x_{ij}|^2.$$

B2. It holds that

$$\min_{i,j \in \mathbb{L}} \left\{ 2v_{ij} + 2k_{ij} - 6\lambda_{ij} - 2a_{ji}P_{ji} - l_{ij} - b_{ji}Q_{ji} - \delta^2 \right\} > \max_{i,j \in \mathbb{L}} \left\{ 5l_{ij} + b_{ji}Q_{ji} \right\} + \max_{i,j \in \mathbb{L}} \left\{ 10s_{ij} + 2c_{ji}^2V_{ji}^2 \right\},$$

where $\lambda_{ij} = \sum_{k=1, k \neq i}^n (a_{jk}P_{jk} + a_{ik}P_{ik})$, $l_{ij} = \sum_{k=1, k \neq i}^n (b_{jk}Q_{jk} + b_{ik}Q_{ik})$, $s_{ij} = \sum_{k=1, k \neq i}^n (c_{jk}^2V_{jk}^2 + c_{ik}^2V_{ik}^2)$, $\eta_{ijk} = \max\{a_{jk}P_{jk} + a_{ik}P_{ik}, b_{jk}Q_{jk} + b_{ik}Q_{ik}, c_{jk}^2V_{jk}^2 + c_{ik}^2V_{ik}^2\}$.

Proof. To begin with, we define

$$V^{ij}(x_{ij}, t) = |x_j - x_i|^2. \quad (24)$$

Obviously, it satisfies condition A1. For $lT \leq t < (l + \theta)T$, we have

$$\begin{aligned} \mathcal{LV}^{ij}(x_{ij}(t), t) &= 2x_{ij}^T(t) \left(f(x_j(t), t) - f(x_i(t), t) + \sum_{k=1}^n a_{jk}H_{jk} - \sum_{k=1}^n a_{ik}H_{ik} + \sum_{k=1}^n b_{jk}M_{jk} - \sum_{k=1}^n b_{ik}M_{ik} + u_{ij}(t) \right) \\ &\quad + \frac{1}{2} \left(g(x_j(t), t) - g(x_i(t), t) + \sum_{k=1}^n c_{jk}N_{jk} - \sum_{k=1}^n c_{ik}N_{ik} \right)^T 2I \left(g(x_j(t), t) - g(x_i(t), t) + \sum_{k=1}^n c_{jk}N_{jk} - \sum_{k=1}^n c_{ik}N_{ik} \right) \\ &\leq -2v_{ij}|x_{ij}(t)|^2 - 2k_{ij}|x_{ij}(t)|^2 + 2x_{ij}^T(t) \left(\sum_{k=1}^n a_{jk}H_{jk} - \sum_{k=1}^n a_{ik}H_{ik} \right) + 2x_{ij}^T(t) \left(\sum_{k=1}^n b_{jk}M_{jk} - \sum_{k=1}^n b_{ik}M_{ik} \right) \\ &\quad + |g(x_j(t), t) - g(x_i(t), t)|^2 + \left| \sum_{k=1}^n c_{jk}N_{jk} - \sum_{k=1}^n c_{ik}N_{ik} \right|^2. \end{aligned} \quad (25)$$

By using (5), it is easy to find that

$$\begin{aligned} &2x_{ij}^T(t) \left(\sum_{k=1}^n a_{jk}H_{jk} - \sum_{k=1}^n a_{ik}H_{ik} \right) \\ &\leq 2|x_{ij}^T(t)| \left(\sum_{k=1, k \neq i}^n a_{jk}|H_{jk}| + \sum_{k=1, k \neq i}^n a_{ik}|H_{ik}| + a_{ji}|H_{ji}| + a_{ii}|H_{ii}| \right) \\ &\leq 2|x_{ij}^T(t)| \sum_{k=1, k \neq i}^n a_{jk}P_{jk}|x_j(t) - x_k(t)| + 2|x_{ij}^T(t)| \sum_{k=1, k \neq i}^n a_{ik}P_{ik}|x_i(t) - x_k(t)| + 2a_{ji}P_{ji}|x_{ij}^T(t)||x_j(t) - x_i(t)| \\ &\leq \sum_{k=1, k \neq i}^n a_{jk}P_{jk}(|x_{ij}(t)|^2 + |x_{jk}(t)|^2) + \sum_{k=1, k \neq i}^n a_{ik}P_{ik}(|x_{ij}(t)|^2 + |x_{ik}(t)|^2) + 2a_{ji}P_{ji}|x_{ij}(t)|^2 \\ &\leq \left(\sum_{k=1, k \neq i}^n (a_{jk}P_{jk} + a_{ik}P_{ik}) + 2a_{ji}P_{ji} \right) |x_{ij}(t)|^2 + \sum_{k=1, k \neq i}^n (a_{jk}P_{jk} + a_{ik}P_{ik}) (|x_{jk}(t)|^2 + |x_{ik}(t)|^2) \\ &\leq (3\lambda_{ij} + 2a_{ji}P_{ji})|x_{ij}(t)|^2 + 3 \sum_{k=1, k \neq i}^n (a_{jk}P_{jk} + a_{ik}P_{ik})|x_{ik}(t)|^2 \\ &= (6\lambda_{ij} + 2a_{ji}P_{ji})|x_{ij}(t)|^2 + 3 \sum_{k=1, k \neq i}^n (a_{jk}P_{jk} + a_{ik}P_{ik}) (|x_{ik}(t)|^2 - |x_{ij}(t)|^2). \end{aligned} \quad (26)$$

Then, according to (6), we can obtain

$$\begin{aligned} &2x_{ij}^T(t) \left(\sum_{k=1}^n b_{jk}M_{jk} - \sum_{k=1}^n b_{ik}M_{ik} \right) \\ &\leq 2|x_{ij}^T(t)| \left(\sum_{k=1, k \neq i}^n b_{jk}|M_{jk}| + \sum_{k=1, k \neq i}^n b_{ik}|M_{ik}| + b_{ji}|M_{ji}| + b_{ii}|M_{ii}| \right) \\ &\leq 2|x_{ij}^T(t)| \left(\sum_{k=1, k \neq i}^n b_{jk}Q_{jk}|x_j(t - \tau_1(t)) - x_k(t - \tau_1(t))| + \sum_{k=1, k \neq i}^n b_{ik}Q_{ik}|x_i(t - \tau_1(t)) - x_k(t - \tau_1(t))| \right. \\ &\quad \left. + b_{ji}Q_{ji}|x_j(t - \tau_1(t)) - x_i(t - \tau_1(t))| \right) \\ &\leq \sum_{k=1, k \neq i}^n b_{jk}Q_{jk}(|x_{ij}(t)|^2 + |x_{jk}(t - \tau_1(t))|^2) + \sum_{k=1, k \neq i}^n b_{ik}Q_{ik}(|x_{ij}(t)|^2 + |x_{ik}(t - \tau_1(t))|^2) \\ &\quad + b_{ji}Q_{ji}(|x_{ij}(t)|^2 + |x_{ij}(t - \tau_1(t))|^2) \\ &\leq \left(\sum_{k=1, k \neq i}^n (b_{jk}Q_{jk} + b_{ik}Q_{ik}) + b_{ji}Q_{ji} \right) |x_{ij}(t)|^2 + \sum_{k=1, k \neq i}^n (b_{jk}Q_{jk} + b_{ik}Q_{ik}) (|x_{jk}(t - \tau_1(t))|^2 + |x_{ik}(t - \tau_1(t))|^2) \\ &\quad + b_{ji}Q_{ji}|x_{ij}(t - \tau_1(t))|^2 \end{aligned}$$

$$\begin{aligned}
&\leq (l_{ij} + b_{ji}Q_{ji})|x_{ij}(t)|^2 + (2l_{ij} + b_{ji}Q_{ji})|x_{ij}(t - \tau_1(t))|^2 + 3 \sum_{k=1, k \neq i}^n (b_{jk}Q_{jk} + b_{ik}Q_{ik})|x_{ik}(t - \tau_1(t))|^2 \\
&= (l_{ij} + b_{ji}Q_{ji})|x_{ij}(t)|^2 + \mu_{ij}|x_{ij}(t - \tau_1(t))|^2 + 3 \sum_{k=1, k \neq i}^n (b_{jk}Q_{jk} + b_{ik}Q_{ik})\left(|x_{ik}(t - \tau_1(t))|^2 - |x_{ij}(t - \tau_1(t))|^2\right), \quad (27)
\end{aligned}$$

where $\mu_{ij} = 5l_{ij} + b_{ji}Q_{ji}$. Form (7), one can get

$$\begin{aligned}
\left| \sum_{k=n}^n c_{jk}N_{jk} - \sum_{k=1}^n c_{ik}N_{ik} \right|^2 &\leq 2 \sum_{k=1, k \neq i}^n c_{jk}^2 |N_{jk}|^2 + 2 \sum_{k=1, k \neq i}^n c_{ik}^2 |N_{ik}|^2 + 2c_{ji}^2 |N_{ji}|^2 + 2c_{ii}^2 |N_{ii}|^2 \\
&\leq 2 \sum_{k=1, k \neq i}^n c_{jk}^2 V_{jk}^2 |x_j(t - \tau_2(t)) - x_k(t - \tau_2(t))|^2 + 2 \sum_{k=1, k \neq i}^n c_{ik}^2 V_{ik}^2 |x_i(t - \tau_2(t)) - x_k(t - \tau_2(t))|^2 \\
&\quad + 2c_{ji}^2 V_{ji}^2 |x_j(t - \tau_2(t)) - x_i(t - \tau_2(t))|^2 \\
&\leq 2 \sum_{k=1, k \neq i}^n (c_{jk}^2 V_{jk}^2 + c_{ik}^2 V_{ik}^2) \left(|x_{jk}(t - \tau_2(t))|^2 + |x_{ik}(t - \tau_2(t))|^2 \right) + 2c_{ji}^2 V_{ji}^2 |x_{ij}(t - \tau_2(t))|^2 \\
&\leq (10\varsigma_{ij} + 2c_{ji}^2 V_{ji}^2) |x_{ij}(t - \tau_2(t))|^2 + 6 \sum_{k=1}^n (c_{jk}^2 V_{jk}^2 + c_{ik}^2 V_{ik}^2) \left(|x_{ik}(t - \tau_2(t))|^2 - |x_{ij}(t - \tau_2(t))|^2 \right) \\
&= \rho_{ij} |x_{ij}(t - \tau_2(t))|^2 + 6 \sum_{k=1, k \neq i}^n (c_{jk}^2 V_{jk}^2 + c_{ik}^2 V_{ik}^2) \left(|x_{ik}(t - \tau_2(t))|^2 - |x_{ij}(t - \tau_2(t))|^2 \right), \quad (28)
\end{aligned}$$

where $\rho_{ij} = 10\varsigma_{ij} + 2c_{ji}^2 V_{ji}^2$. Substituting (18), (26)–(28) into (25), we can obtain that

$$\begin{aligned}
\mathcal{L}V^{ij}(x_{ij}(t), t) &\leq \left(-2v_{ij} - 2k_{ij} + 6\lambda_{ij} + 2a_{ji}P_{ji} + l_{ij} + b_{ji}Q_{ji} + \delta^2 \right) |x_{ij}(t)|^2 + \mu_{ij}|x_{ij}(t - \tau_1(t))|^2 + \rho_{ij}|x_{ij}(t - \tau_2(t))|^2 \\
&\quad + 3 \sum_{k=1, k \neq i}^n (a_{jk}P_{jk} + a_{ik}P_{ik}) \left(|x_{ik}(t)|^2 - |x_{ij}(t)|^2 \right) \\
&\quad + 3 \sum_{k=1, k \neq i}^n (b_{jk}Q_{jk} + b_{ik}Q_{ik}) \left(|x_{ik}(t - \tau_1(t))|^2 - |x_{ij}(t - \tau_1(t))|^2 \right) \\
&\quad + 6 \sum_{k=1, k \neq i}^n (c_{jk}^2 V_{jk}^2 + c_{ik}^2 V_{ik}^2) \left(|x_{ik}(t - \tau_2(t))|^2 - |x_{ij}(t - \tau_2(t))|^2 \right) \\
&\leq -\sigma_{ij}^{(1)} |x_{ij}(t)|^2 + \mu_{ij}|x_{ij}(t - \tau_1(t))|^2 + \rho_{ij}|x_{ij}(t - \tau_2(t))|^2 \\
&\quad + 3 \sum_{k=1, k \neq i}^n \eta_{ijk} \left(|x_{ik}(t)|^2 + |x_{ik}(t - \tau_1(t))|^2 + 2|x_{ik}(t - \tau_2(t))|^2 \right. \\
&\quad \left. - \left(|x_{ij}(t)|^2 + |x_{ij}(t - \tau_1(t))|^2 + 2|x_{ij}(t - \tau_2(t))|^2 \right) \right) \\
&\leq -\sigma_{ij}^{(1)} |x_{ij}(t)|^2 + \mu_{ij}|x_{ij}(t - \tau_1(t))|^2 + \rho_{ij}|x_{ij}(t - \tau_2(t))|^2 + \sum_{k=1, k \neq i}^n h_{ijk} \left(G_{ik}(X_{ik}(t)) - G_{ij}(X_{ij}(t)) \right),
\end{aligned}$$

in which

$$\begin{aligned}
\sigma_{ij}^{(1)} &= 2v_{ij} + 2k_{ij} - 6\lambda_{ij} - 2a_{ji}P_{ji} - l_{ij} - b_{ji}Q_{ji} - \delta^2, \\
G_{ik}(X_{ik}(t)) &= 3 \left(|x_{ik}(t)|^2 + |x_{ik}(t - \tau_1(t))|^2 + 2|x_{ik}(t - \tau_2(t))|^2 \right), \\
G_{ij}(X_{ij}(t)) &= 3 \left(|x_{ij}(t)|^2 + |x_{ij}(t - \tau_1(t))|^2 + 2|x_{ij}(t - \tau_2(t))|^2 \right)
\end{aligned}$$

and

$$h_{ijk} = \begin{cases} \eta_{ijk}, & j \neq k, \\ 0, & j = k. \end{cases}$$

Similarly, for $(l + \theta)t \leq t < (l + 1)t$, we have

$$\begin{aligned} \mathcal{L}V^{ij}(x_{ij}(t), t) &= 2x_{ij}^T(t) \left(f(x_j(t), t) - f(x_i(t), t) + \sum_{k=1}^n a_{jk}H_{jk} - \sum_{k=1}^n a_{ik}H_{ik} + \sum_{k=1}^n b_{jk}M_{jk} - \sum_{k=1}^n b_{ik}M_{ik} \right) \\ &\quad + \frac{1}{2} \left(g(x_j(t), t) - g(x_i(t), t) + \sum_{k=1}^n c_{jk}N_{jk} - \sum_{k=1}^n c_{ik}N_{ik} \right)^T 2I \\ &\quad \times \left(g(x_j(t), t) - g(x_i(t), t) + \sum_{k=1}^n c_{jk}N_{jk} - \sum_{k=1}^n c_{ik}N_{ik} \right) \\ &\leq \left(-2v_{ij} + 6\lambda_{ij} + 2a_{ji}P_{ji} + l_{ij} + b_{ji}Q_{ji} + \delta^2 \right) |x_{ij}(t)|^2 + \mu_{ij}|x_{ij}(t - \tau_1(t))|^2 + \rho_{ij}|x_{ij}(t - \tau_2(t))|^2 \\ &\quad + 3 \sum_{k=1, k \neq i}^n \left(a_{jk}P_{jk} + a_{ik}P_{ik} \right) \left(|x_{ik}(t)|^2 - |x_{ij}(t)|^2 \right) \\ &\quad + 3 \sum_{k=1, k \neq i}^n \left(b_{jk}Q_{jk} + b_{ik}Q_{ik} \right) \left(|x_{ik}(t - \tau_1(t))|^2 - |x_{ij}(t - \tau_1(t))|^2 \right) \\ &\quad + 6 \sum_{k=1, k \neq i}^n \left(c_{jk}^2V_{jk}^2 + c_{ik}^2V_{ik}^2 \right) \left(|x_{ik}(t - \tau_2(t))|^2 - |x_{ij}(t - \tau_2(t))|^2 \right) \\ &\leq -\sigma_{ij}^{(2)} |x_{ij}(t)|^2 + \mu_{ij}|x_{ij}(t - \tau_1(t))|^2 + \rho_{ij}|x_{ij}(t - \tau_2(t))|^2 + \sum_{k=1, k \neq i}^n h_{ijk} \left(G_{ik}(X_{ik}(t)) - G_{ij}(X_{ij}(t)) \right), \end{aligned}$$

where

$$\sigma_{ij}^{(2)} = 2v_{ij} - 6\lambda_{ij} - 2a_{ji}P_{ji} - l_{ij} - b_{ji}Q_{ji} - \delta^2.$$

Above all, we have verified that $\mathcal{L}V^{ij}(x_{ij}(t), t)$ satisfies condition A2. Recalling to Theorem 1, Lemma 2 and condition B2, we can conclude that system (1) can achieve IES under periodically intermittent control (3). This completes the proof. $\square$

Remark 5. Based on a relatively general result given in the form of Lyapunov-type theorem, we take a further step to derive a coefficients-type theorem whose conditions mainly depend on the coefficients in system (4). It is clear that the coefficient-type theorem is convenient to check and quite useful in practice. These points will be shown by an application to stochastic coupled oscillator with time-varying delays in the next section.

4. An application to stochastic coupled oscillators with time-varying delays

Synchronization of oscillators model is extensively studied in physical and biological systems for underlying interests ranging from novel communications strategies to understand how large and small neural assemblies efficiently and sensitively achieve desired functional goals. In this section, we will apply the main results to discuss IES of stochastic coupled oscillators with time-varying delays. Firstly, a common oscillator reads

$$\ddot{x}(t) + \varphi \dot{x}(t) + x(t) = 0, \quad t \geq 0, \quad (29)$$

where $\varphi \geq 0$ is the damping coefficient.

It has been observed that stochastic perturbations and time-varying delays are almost inevitable in real systems. Then, we consider a digraph $\mathcal{G}$ with n ($n \geq 2$) vertices, each vertex is assigned a stochastic coupled oscillator described by (29). Combining with the coupling, time-varying delays and white noise effect, we get the stochastic coupled oscillators with time-varying delays of the form

$$\begin{aligned} \ddot{x}_i(t) + \varphi \dot{x}_i(t) + x_i(t) &+ \sum_{k=1}^n d_{ik}F_{ik}(x_i(t), x_k(t)) + \sum_{k=1}^n h_{ik}Y_{ik}(x_i(t - \tau_1(t)), x_k(t - \tau_1(t))) + m(t) \\ &= \left(g(x_i(t), t) + \sum_{k=1}^n m_{ik}Z_{ik}(x_i(t - \tau_2(t)), x_k(t - \tau_2(t))) \right) \dot{W}(t), \quad t \geq 0, \quad i \in \mathbb{L}, \end{aligned} \quad (30)$$

where $x_i(t) \in \mathbb{R}$ is the state variant of the i th vertex system; g stands for the perturbation intense of white noise; d_{ik}, h_{ik}, m_{ik} denote the intension of influence from k th vertex to i th vertex. It is easy to find that $d_{ik}, h_{ik}, m_{ik} \geq 0$, and if and only if no arc from vertex k to i in $\mathcal{G}$, $d_{ik}, h_{ik}, m_{ik} = 0$. Meanwhile, $F_{ik}, Y_{ik} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ denote the influence of the state of k th vertex on i th vertex with or without delay, and $Z_{ik} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ represents the influence of the intension of white noise g of the k th vertex on the i th vertex.

Hereinafter, for any $i, k \in \mathbb{L}$ denote

$$F_{ik} = F_{ik}(x_i(t), x_k(t)), \quad Y_{ik} = Y_{ik}(x_i(t - \tau_1(t)), x_k(t - \tau_1(t))), \quad Z_{ik} = Z_{ik}(x_i(t - \tau_2(t)), x_k(t - \tau_2(t))).$$

By making a transformation of $y_i(t) = \dot{x}_i(t) + \varrho x_i(t)$, where $\varrho > 0$ is a constant, then system (30) is equivalent to

$$\begin{cases} dx_i(t) = (y_i(t) - \varrho x_i(t)) dt, \\ dy_i(t) = ((\varrho - \varphi)y_i(t) + (\varrho\varphi - \varrho^2 - 1)x_i(t) - \sum_{k=1}^n d_{ik}F_{ik} - \sum_{k=1}^n h_{ik}Y_{ik} - m(t)) dt \quad t \geq 0, \quad i \in \mathbb{L}, \\ \quad + (g(x_i(t), t) + \sum_{k=1}^n m_{ik}Z_{ik}) dW(t), \end{cases} \quad (31)$$

Here, we continue to use periodically intermittent control strategy to drive system (30) achieve IES. For any fixed $i \in \mathbb{L}$, periodically intermittent controls are imposed on the j th ($j \in \mathbb{L}, j \neq i$) vertex system. Then the j th vertex system with periodically intermittent control can be described by

$$\begin{cases} dx_j(t) = (y_j(t) - \varrho x_j(t) - u_{ij}^{(1)}(t)) dt, \\ dy_j(t) = ((\varrho - \varphi)y_j(t) + (\varrho\varphi - \varrho^2 - 1)x_j(t) - \sum_{k=1}^n d_{jk}F_{jk} - \sum_{k=1}^n h_{jk}Y_{jk} - m(t) - u_{ij}^{(2)}(t)) dt \quad t \geq 0, \\ \quad + (g(x_j(t), t) + \sum_{k=1}^n m_{jk}Z_{jk}) dW(t), \end{cases} \quad (32)$$

where $u_{ij}^{(1)}(t)$ and $u_{ij}^{(2)}(t)$ are periodically intermittent controls of the following forms:

$$u_{ij}^{(1)}(t) = \begin{cases} -k_{ij}^{(1)}(x_j(t) - x_i(t)), & lT \leq t < (l + \theta)T, \\ 0, & (l + \theta)T \leq t < (l + 1)T \end{cases} \quad (33)$$

and

$$u_{ij}^{(2)}(t) = \begin{cases} -k_{ij}^{(2)}(y_j(t) - y_i(t)), & lT \leq t < (l + \theta)T, \\ 0, & (l + \theta)T \leq t < (l + 1)T, \end{cases} \quad (34)$$

in which $l \in \mathbb{Z}^+$, $k_{ij}^{(1)} > 0$ and $k_{ij}^{(2)} > 0$ denote the control strengths, T is the control period, and $0 < \theta < 1$ is the rate of control duration.

For convenience, we denote

$$e_i(t) = \begin{pmatrix} x_i(t) \\ y_i(t) \end{pmatrix}, \quad e_i(t) = \begin{pmatrix} y_i(t) - \varrho x_i(t) \\ (\varrho - \varphi)y_i(t) + (\varrho\varphi - \varrho^2 - 1)x_i(t) \end{pmatrix},$$

$$e_i(t) = \begin{pmatrix} 0 \\ -F_{ik} \end{pmatrix}, \quad e_i(t) = \begin{pmatrix} 0 \\ -Y_{ik} \end{pmatrix}, \quad e_i(t) = \begin{pmatrix} 0 \\ Z_{ik} \end{pmatrix}, \quad e_i(t) = \begin{pmatrix} 0 \\ g(x_i(t), t) \end{pmatrix}.$$

Hence, system (31) and (32) can be rewritten as

$$de_i(t) = \left(f(e_i(t), t) + \sum_{k=1}^n d_{ik}H_{ik} + \sum_{k=1}^n h_{ik}M_{ik} + I(t) \right) dt + \left(g(e_i(t), t) + \sum_{k=1}^n m_{ik}N_{ik} \right) dW(t), \quad t \geq 0, \quad i \in \mathbb{L} \quad (35)$$

and

$$de_j(t) = \left(f(e_j(t), t) + \sum_{k=1}^n d_{jk}H_{jk} + \sum_{k=1}^n h_{jk}M_{jk} + I(t) + U_{ij}(t) \right) dt + \left(g(e_j(t), t) + \sum_{k=1}^n m_{jk}N_{jk} \right) dW(t), \quad t \geq 0, \quad j \neq i.$$

Define $e_{ij}(t) = e_j(t) - e_i(t)$ as the inner synchronization error, then the error system can be expressed as

$$\begin{aligned} de_{ij}(t) = & \left(f(e_{ij}(t), t) + \sum_{k=1}^n d_{jk}H_{jk} - \sum_{k=1}^n d_{ik}H_{ik} + \sum_{k=1}^n h_{jk}M_{jk} - \sum_{k=1}^n h_{ik}M_{ik} + U_{ij}(t) \right) dt \\ & + \left(g(e_{ij}(t), t) + \sum_{k=1}^n m_{jk}N_{jk} - \sum_{k=1}^n m_{ik}N_{ik} \right) dW(t), \quad t \geq 0, \quad i, j \in \mathbb{L}, \quad j \neq i, \end{aligned} \quad (36)$$

where $f(e_{ij}(t), t) = f(e_j(t), t) - f(e_i(t), t)$ and $g(e_{ij}(t), t) = g(e_j(t), t) - g(e_i(t), t)$.

Afterwards, a theorem is given to guarantee IES of system (35) under periodically intermittent controls (33) and (34).

Theorem 3. For any fixed i ($i \in \mathbb{L}$) and any j ($j \in \mathbb{L}, j \neq i$), let diagraph $(\mathcal{G}, R)$ be strongly connected. System (35) is inner exponentially synchronized under periodically intermittent controls (33) and (34) if the following conditions hold:

C1. There exist positive scalars J_{ik}, K_{ik} and L_{ik} such that

$$|F_{ik}(x_i, x_k)| \leq J_{ik}|x_i - x_k|, \quad |Y_{ik}(x_i, x_k)| \leq K_{ik}|x_i - x_k|, \quad \text{and} \quad |Z_{ik}(x_i, x_k)| \leq L_{ik}|x_i - x_k|.$$

C2. There is a positive constant ζ such that

$$|g(x_i, t) - g(x_j, t)| \leq \zeta |x_i - x_j|.$$

C3. It holds that

$$\min_{i,j \in \mathbb{L}} \left\{ \frac{\varphi}{2} + 2k_{ij} - 6\psi_{ij} - 2d_{ij}J_{ij} - \chi_{ij} - h_{ij}K_{ij} - \zeta^2 \right\} > \max_{i,j \in \mathbb{L}} \left\{ 5\chi_{ij} + h_{ij}K_{ij} \right\} + \max_{i,j \in \mathbb{L}} \left\{ 10\varpi_{ij} + 2m_{ij}^2 L_{ij}^2 \right\},$$

where $\psi_{ij} = \sum_{k=1, k \neq i}^n (d_{jk}J_{jk} + d_{ik}J_{ik})$, $\chi_{ij} = \sum_{k=1, k \neq i}^n (h_{jk}K_{jk} + h_{ik}K_{ik})$, $\varpi_{ij} = \sum_{k=1, k \neq i}^n (m_{jk}^2 L_{jk}^2 + m_{ik}^2 L_{ik}^2)$, $k_{ij} = \min\{k_{ij}^{(1)}, k_{ij}^{(2)}\}$, $y_{ijk} = \max\{d_{jk}J_{jk} + d_{ik}J_{ik}, h_{jk}K_{jk} + h_{ik}K_{ik}, m_{jk}^2 L_{jk}^2 + m_{ik}^2 L_{ik}^2\}$, $R = (r_{ijk})_{(n-1) \times (n-1)}$ and

$$r_{ijk} = \begin{cases} y_{ijk}, & j \neq k, \\ 0, & j = k. \end{cases}$$

Proof. For any $\varphi > 4$, we can derive that there exists a constant $\varrho > 0$ such that

$$\omega_1 = -\frac{|\varrho\varphi - \varrho^2| - 2\varrho}{2} > \varphi/2, \quad \omega_2 = -\frac{|\varrho\varphi - \varrho^2| + 2(\varrho - \varphi)}{2} > \varphi/2.$$

Let $y_{ij} = y_j - y_i$, then we find that

$$\begin{aligned} e_{ij}^T f(e_{ij}, t) &= \begin{pmatrix} x_{ij} \\ y_{ij} \end{pmatrix}^T \begin{pmatrix} y_{ij} - \varrho x_{ij} \\ (\varrho - \varphi)y_{ij} + (\varrho\varphi - \varrho^2 - 1)x_{ij} \end{pmatrix} \\ &= x_{ij}y_{ij} - \varrho|x_{ij}|^2 + (\varrho - \varphi)|y_{ij}|^2 + (\varrho\varphi - \varrho^2 - 1)x_{ij}y_{ij} \\ &\leq -\varrho|x_{ij}|^2 + (\varrho - \varphi)|y_{ij}|^2 + \frac{1}{2}|\varrho\varphi - \varrho^2|(|x_{ij}|^2 + |y_{ij}|^2) \\ &= -\omega_1|x_{ij}|^2 - \omega_2|y_{ij}|^2 \\ &\leq -\omega|e_{ij}|^2, \end{aligned}$$

where $\omega = \min\{\omega_1, \omega_2\}$. Therefore $\omega > \varphi/2$. It follows from condition C3 that:

$$\begin{aligned} &\min_{i,j \in \mathbb{L}} \left\{ \omega + 2k_{ij} - 6\psi_{ij} - 2d_{ij}J_{ij} - \chi_{ij} - h_{ij}K_{ij} - \zeta^2 \right\} \\ &> \min_{i,j \in \mathbb{L}} \left\{ \varphi/2 + 2k_{ij} - 6\psi_{ij} - 2d_{ij}J_{ij} - \chi_{ij} - h_{ij}K_{ij} - \zeta^2 \right\} \\ &> \max_{i,j \in \mathbb{L}} \left\{ 5\chi_{ij} + h_{ij}K_{ij} \right\} + \max_{i,j \in \mathbb{L}} \left\{ 10\varpi_{ij} + 2m_{ij}^2 L_{ij}^2 \right\}. \end{aligned}$$

Then, we have

$$\min_{i,j \in \mathbb{L}} \left\{ 2\nu_{ij} + 2k_{ij} - 6\lambda_{ij} - 2a_{ji}P_{ji} - l_{ij} - b_{ji}Q_{ji} - \delta^2 \right\} > \max_{i,j \in \mathbb{L}} \left\{ 5l_{ij} + b_{ji}Q_{ji} \right\} + \min_{i,j \in \mathbb{L}} \left\{ 10\varsigma_{ij} + 2c_{ij}^2 V_{ij}^2 \right\},$$

where $\omega = 2\nu_{ij}$, $\psi_{ij} = \lambda_{ij}$, $\chi_{ij} = l_{ij}$, $\varpi_{ij} = \varsigma_{ij}$. Hence, condition B2 holds.

Furthermore, from C1 and C2, we obtain that

$$\begin{aligned} |H_{ik}(e_i, e_k)|^2 &= \left| \begin{pmatrix} 0 \\ -F_{ik}(x_i, x_k) \end{pmatrix} \right|^2 = |-F_{ik}(x_i, x_k)|^2 \leq J_{jk}|x_{jk}|^2 \leq J_{jk}|e_{jk}|^2, \\ |M_{ik}(e_i, e_k)|^2 &= \left| \begin{pmatrix} 0 \\ -Y_{ik}(x_i, x_k) \end{pmatrix} \right|^2 = |-Y_{ik}(x_i, x_k)|^2 \leq K_{jk}|x_{jk}|^2 \leq K_{jk}|e_{jk}|^2, \\ |N_{ik}(e_i, e_k)|^2 &= \left| \begin{pmatrix} 0 \\ Z_{ik}(x_i, x_k) \end{pmatrix} \right|^2 = |Z_{ik}(x_i, x_k)|^2 \leq L_{jk}|x_{jk}|^2 \leq L_{jk}|e_{jk}|^2 \end{aligned}$$

and

$$|g(e_{ij}, t)|^2 = \left| \begin{pmatrix} 0 \\ g(x_j, t) - g(x_i, t) \end{pmatrix} \right|^2 = |g(x_j, t) - g(x_i, t)|^2 \leq \zeta |x_{ij}|^2 \leq \zeta |e_{ij}|^2.$$

Thus, all the conditions in Theorem 2 are fulfilled. That is to say, system (35) can realize IES under periodically intermittent controls (33) and (34). $\square$

Remark 6. Up to now, the dynamical behaviors of coupled oscillators with or without stochastic perturbations and time-varying delays have been widely studied and some works are reported on their stability [34], synchronization [36]. In [34], the authors considered global exponential stability for stochastic coupled oscillators with variable delay by using the graph-theoretical method. To our knowledge, there are few results on IES analysis of stochastic coupled oscillators with time-varying delays based on the graph-theoretical method. Moreover, periodically intermittent control is designed to synchronize the system (30). So our results endow stochastic coupled oscillators with time-varying delays more applicability in physics and other fields.

Remark 7. In Theorem 3, some sufficient criteria are given to ensure IES of system (30). Recently, some sufficient synchronization laws of stochastic coupled oscillators have been established [28,36]. It should be pointed that these criteria are influenced by transformation parameter ϱ in the transformation of $y_i(t) = \dot{x}_i(t) + \varrho x_i(t)$. As a matter of fact, the dynamical behaviors of stochastic coupled oscillators system (30) should not be affected by the transformation parameters. The IES criteria derived in this paper are only dependent on the coefficients in system (30) and unrelated to the transformation parameter ϱ .

5. Numerical Test

In this section, we provide an example with numerical simulation to demonstrate the effectiveness of the proposed theoretical results.

Here we proceed to consider the IES problem of stochastic coupled oscillators (30) under periodically intermittent control. Take $n = 5$, system (30) can be expressed by a digraph $\mathcal{G}$, and the coupling strengths d_{ik} , h_{ik} and m_{ik} are shown as follows:

$$(d_{ik})_{(5 \times 5)} = \begin{pmatrix} 0 & 0.0215 & 0.0315 & 0.0400 & 0.0185 \\ 0.0155 & 0 & 0.0115 & 0.0065 & 0.0025 \\ 0.0050 & 0.0100 & 0 & 0 & 0.0012 \\ 0.0105 & 0.0300 & 0.0030 & 0 & 0.0125 \\ 0.0250 & 0.1000 & 0.0015 & 0.0050 & 0 \end{pmatrix},$$

$$(h_{ik})_{(5 \times 5)} = \begin{pmatrix} 0 & 0.0155 & 0.0050 & 0.0200 & 0.0250 \\ 0.0215 & 0 & 0.0100 & 0.0300 & 0.1000 \\ 0.0315 & 0.0115 & 0 & 0.0030 & 0.0015 \\ 0.0114 & 0.0065 & 0 & 0 & 0.0050 \\ 0.0185 & 0.0025 & 0.0012 & 0.0125 & 0 \end{pmatrix},$$

$$(m_{ik})_{(5 \times 5)} = \begin{pmatrix} 0 & 0.0185 & 0.0182 & 0.0300 & 0.0217 \\ 0.0185 & 0 & 0.0107 & 0.0182 & 0.0513 \\ 0.0182 & 0.0107 & 0 & 0.0015 & 0.0013 \\ 0.0114 & 0.0182 & 0.0015 & 0 & 0.0088 \\ 0.0217 & 0.0513 & 0.0013 & 0.0088 & 0 \end{pmatrix}.$$

Furthermore, suppose that $H_{ik}(x_i, x_k) = 0.5 \sin(x_i - x_k)$, $M_{ij}(x_i, x_k) = \cos(x_i) - \cos(x_k)$, $N_{ik}(x_i, x_k) = \tanh(x_i) - \tanh(x_k)$, $\tau_1(t) = 0.5 + 0.5 \sin(t)$, $\tau_2(t) = 0.3 - 0.2 \cos(t)$, $m(t) = 28 \cos(\pi t)$, $g(x_i(t), t) = 0.8 \sin(x_i(t)) + 0.2t$, where $i, j, k = 1, 2, 3, 4, 5$.

For simplicity, we fixed $i = 1$. Other parameters and the control strengths are chosen as $\varrho = 3.4$, $\varphi = 4.1$,

$$k_{1j}^{(1)} = \begin{cases} 11 & j = 2, \\ 12 & j = 3, \\ 13 & j = 4, \\ 14 & j = 5, \end{cases} \quad \text{and} \quad k_{1j}^{(2)} = \begin{cases} 14 & j = 2, \\ 5 & j = 3, \\ 8 & j = 4, \\ 11 & j = 5. \end{cases}$$

Then, by simple computation we can derive that $\omega = 2.21$, $J_{ij} = K_{ij} = L_{ij} = \zeta = 1$, $k_{12} = 14$, $k_{13} = 12$, $k_{14} = 13$ and $k_{15} = 14$ and the weighted matrix of digraph $(\mathcal{G}, R)$ is

$$R = (r_{1jk})_{4 \times 4} = \begin{pmatrix} 0 & 0.3400 & 0.3440 & 0.3905 \\ 0.3400 & 0 & 0.3046 & 0.3706 \\ 0.3440 & 0.3046 & 0 & 0.4200 \\ 0.3905 & 0.3706 & 0.4200 & 0 \end{pmatrix}.$$

Additionally, we can also get that

$$\psi_{1j} = \begin{cases} 0.227 & j = 2, \\ 0.113 & j = 3, \\ 0.117 & j = 4, \\ 0.1002 & j = 5, \end{cases} \quad \chi_{1j} = \begin{cases} 0.1475 & j = 2, \\ 0.1276 & j = 3, \\ 0.177 & j = 4, \\ 0.243 & j = 5 \end{cases} \quad \text{and} \quad \varpi_{1j} = \begin{cases} 0.1873 & j = 2, \\ 0.1203 & j = 3, \\ 0.1470 & j = 4, \\ 0.1716 & j = 5. \end{cases}$$

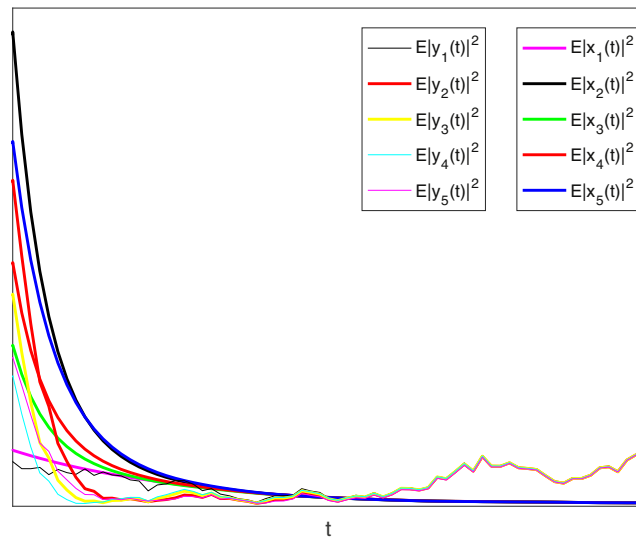


Fig. 1. The paths of mean square of the solution to system (30) under periodically intermittent controls (33) and (34).

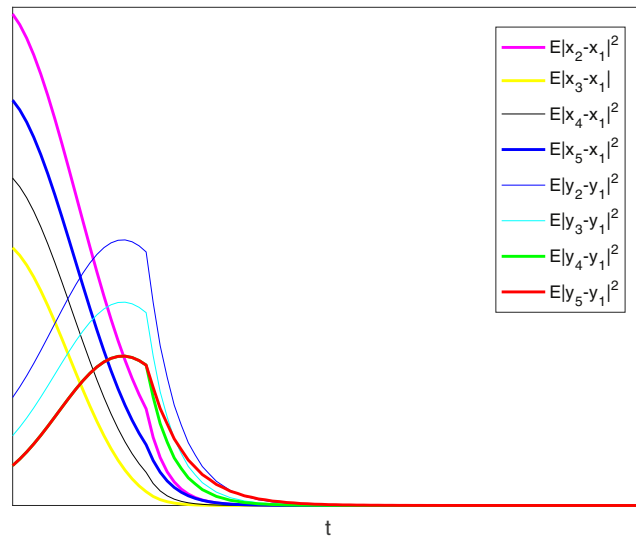


Fig. 2. The responses of mean square of errors $x_j(t) - x_i(t)$ and $y_j(t) - y_i(t)$, $j = 2, 3, 4, 5$.

What is more, it is straightforward to check that

$$\omega + 2k_{1j} - 6\psi_{1j} - 2d_{1j}l_{1j} - \chi_{1j} - h_{1j}K_{1j} - \zeta^2 = 22.7847, \quad j = 2,$$

$$\omega + 2k_{1j} - 6\psi_{1j} - 2d_{1j}l_{1j} - \chi_{1j} - h_{1j}K_{1j} - \zeta^2 = 25.7172, \quad j = 3,$$

$$\omega + 2k_{1j} - 6\psi_{1j} - 2d_{1j}l_{1j} - \chi_{1j} - h_{1j}K_{1j} - \zeta^2 = 27.6267, \quad j = 4,$$

$$\omega + 2k_{1j} - 6\psi_{1j} - 2d_{1j}l_{1j} - \chi_{1j} - h_{1j}K_{1j} - \zeta^2 = 29.7251, \quad j = 5$$

and

$$5\chi_j + h_{1j}K_{1j} = 0.7711, \quad 10\varpi_{1j} + 2m_{1j}^2L_{1j}^2 = 0.9836, \quad j = 2,$$

$$5\chi_j + h_{1j}K_{1j} = 0.3977, \quad 10\varpi_{1j} + 2m_{1j}^2L_{1j}^2 = 0.8513, \quad j = 3,$$

$$5\chi_j + h_{1j}K_{1j} = 0.4167, \quad 10\varpi_{1j} + 2m_{1j}^2L_{1j}^2 = 1.1808, \quad j = 4,$$

$$5\chi_j + h_{1j}K_{1j} = 0.3462, \quad 10\varpi_{1j} + 2m_{1j}^2L_{1j}^2 = 1.6204, \quad j = 5.$$

Select $\xi = 2.3$, one can derive $\theta > 0.9179$. Hence, all the conditions in Theorem 3 are fulfilled, which implies that system (30) can achieve IES under periodically intermittent controls (33) and (34). For initial conditions, $(90 \sin(0.1t), -20 \cos(0.13t), -70, 60, -70, 50, -50, 40, -60, 40)^T$, the corresponding numerical simulations are shown in Figs. 1, 2 which verify the effectiveness of our theoretical results.

Remark 8. Recently, great effort has been devoted to investigating the synchronization issue of stochastic complex networks with time delays by applying linear matrix inequality (LMI) method. In [37], synchronization of stochastic genetic oscillator networks with time delays and Markovian jumping parameters was studied by using the LMI technique. It should be noticed that they require the help of MATLAB LMI Toolbox to check the sufficient synchronization conditions. In this paper, the exponential synchronization criteria are given in the form of the coefficients systems. It can be clearly seen from Theorem 3 that one can use the coefficients in stochastic coupled oscillators (30) to directly verify the synchronization criteria, which means that the method presented in this paper is more convenient to use in practice.

6. Conclusion

In this paper, we have studied IES of SCSTDNs via periodically intermittent control. By combining graph theory with Lyapunov method, several simple yet generic sufficient conditions for IES of SCSTDNs under periodically intermittent control have been derived. As illustrations, the IES analysis of stochastic coupled oscillators with time-varying delays has been carried out by applying the derived theoretic results. A numerical simulation has been given to show the effectiveness of the theoretical outcomes. Note that, here we only focus on IES of SCSTDNs via periodically intermittent control. However, in reality, particularly in some engineering fields, it is desirable and more valuable that inner synchronization of stochastic coupled systems with or without time-varying delays can be achieved in finite time rather than infinite time. On the other hand, the requirement of periodicity of intermittent control is quite restricted and may unreasonable and unnecessary in practice. Hereinafter, our future research topics including finite-time inner synchronization of coupled system on networks under aperiodically intermittent control and synchronization of stochastic coupled networks via aperiodically intermittent pinning control.

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